

EXP NO: 9

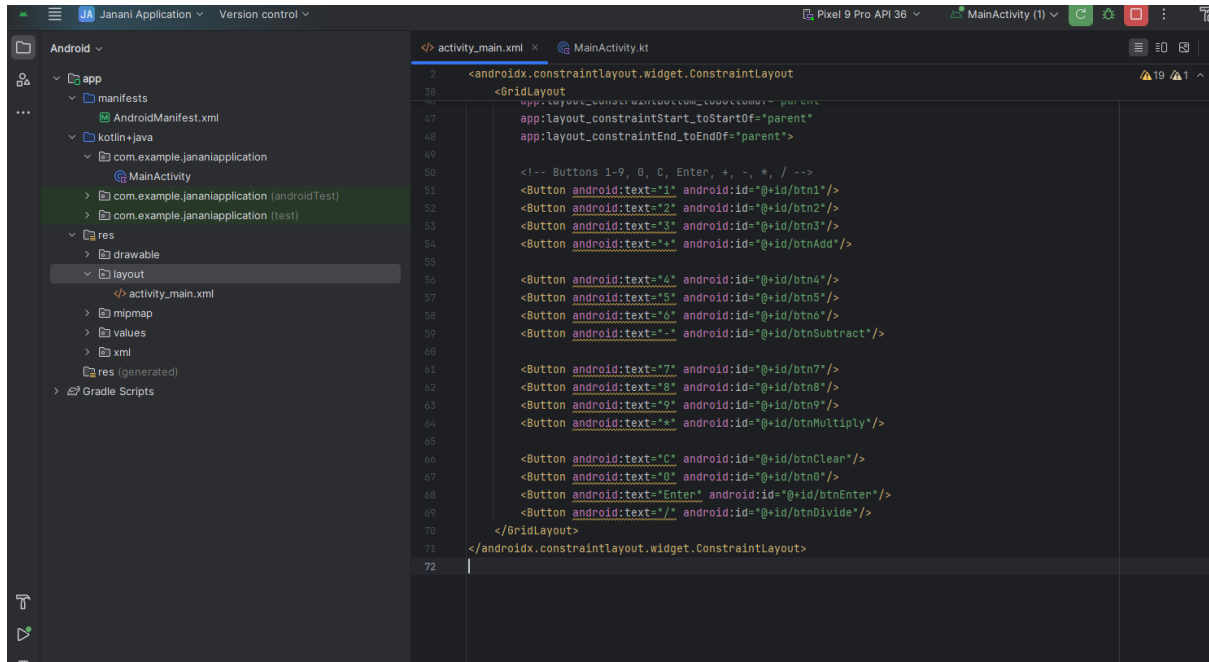
DATE:

DEVELOP AN ANDROID APPLICATION USING CONTROLS LIKE BUTTON,TEXT VIEW,EDIT TEXT FOR CALCULATOR HAVING FUNCTIONALITY ADDITION,SUBTRACTION,MULTIPLICATION AND DIVISION.

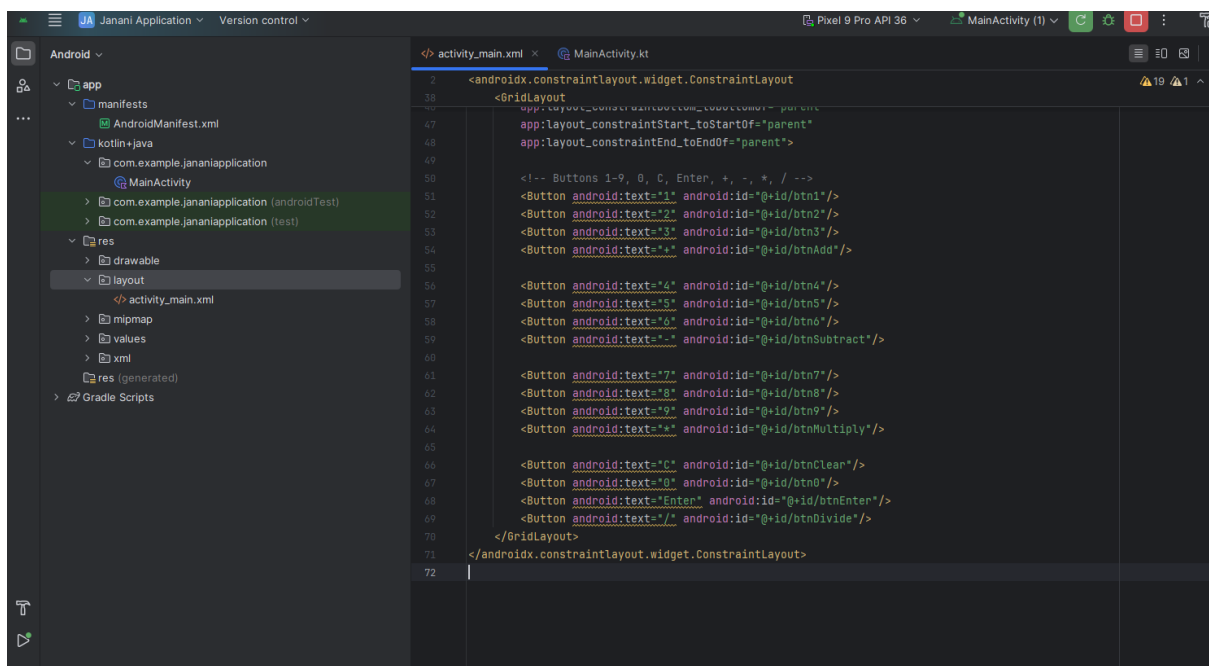
AIM:

To develop a android application using controls like button,textView and edit text for designing a calculator having the basic functionalities like addition,subtraction,multiplication and division.

PROGRAM AND OUTPUT:



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <androidx.constraintlayout.widget.ConstraintLayout
3     xmlns:android="http://schemas.android.com/apk/res/android"
4     xmlns:tools="http://schemas.android.com/tools"
5     android:layout_width="match_parent"
6     android:layout_height="match_parent"
7     tools:context=".MainActivity">
8     <!-- Buttons 1-9, 0, C, Enter, +, -, *, / -->
9     <Button android:text="1" android:id="@+id/btn1"/>
10    <Button android:text="2" android:id="@+id/btn2"/>
11    <Button android:text="3" android:id="@+id/btn3"/>
12    <Button android:text="4" android:id="@+id/btn4"/>
13    <Button android:text="5" android:id="@+id/btn5"/>
14    <Button android:text="6" android:id="@+id/btn6"/>
15    <Button android:text="7" android:id="@+id/btn7"/>
16    <Button android:text="8" android:id="@+id/btn8"/>
17    <Button android:text="9" android:id="@+id/btn9"/>
18    <Button android:text="0" android:id="@+id/btn0"/>
19    <Button android:text="C" android:id="@+id/btnClear"/>
20    <Button android:text="+" android:id="@+id/btnAdd"/>
21    <Button android:text="-" android:id="@+id/btnSubtract"/>
22    <Button android:text="*" android:id="@+id/btnMultiply"/>
23    <Button android:text="/" android:id="@+id/btnDivide"/>
24    <Button android:text="Enter" android:id="@+id/btnEnter"/>
25    </Button>
26    </androidx.constraintlayout.widget.ConstraintLayout>
```



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16    <Button android:text="8" android:id="@+id/btn8"/>
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18    <Button android:text="0" android:id="@+id/btn0"/>
19    <Button android:text="C" android:id="@+id/btnClear"/>
20    <Button android:text="+" android:id="@+id/btnAdd"/>
21    <Button android:text="-" android:id="@+id/btnSubtract"/>
22    <Button android:text="*" android:id="@+id/btnMultiply"/>
23    <Button android:text="/" android:id="@+id/btnDivide"/>
24    <Button android:text="Enter" android:id="@+id/btnEnter"/>
25    </Button>
26    </androidx.constraintlayout.widget.ConstraintLayout>
```

```

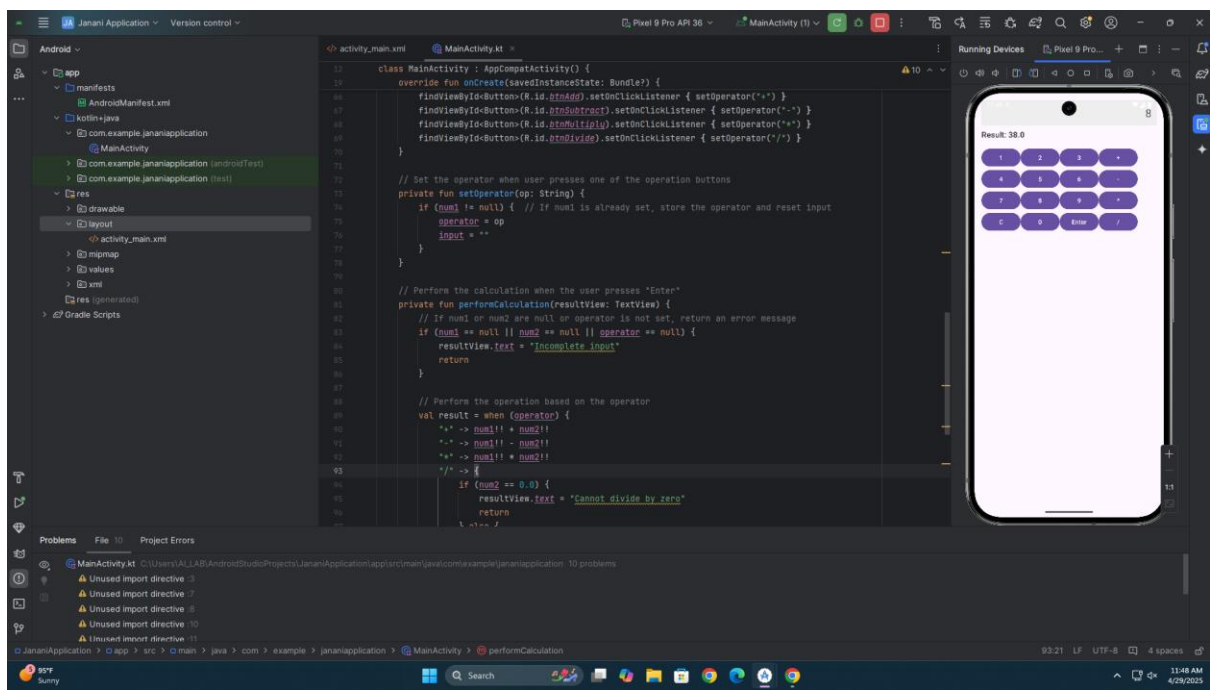
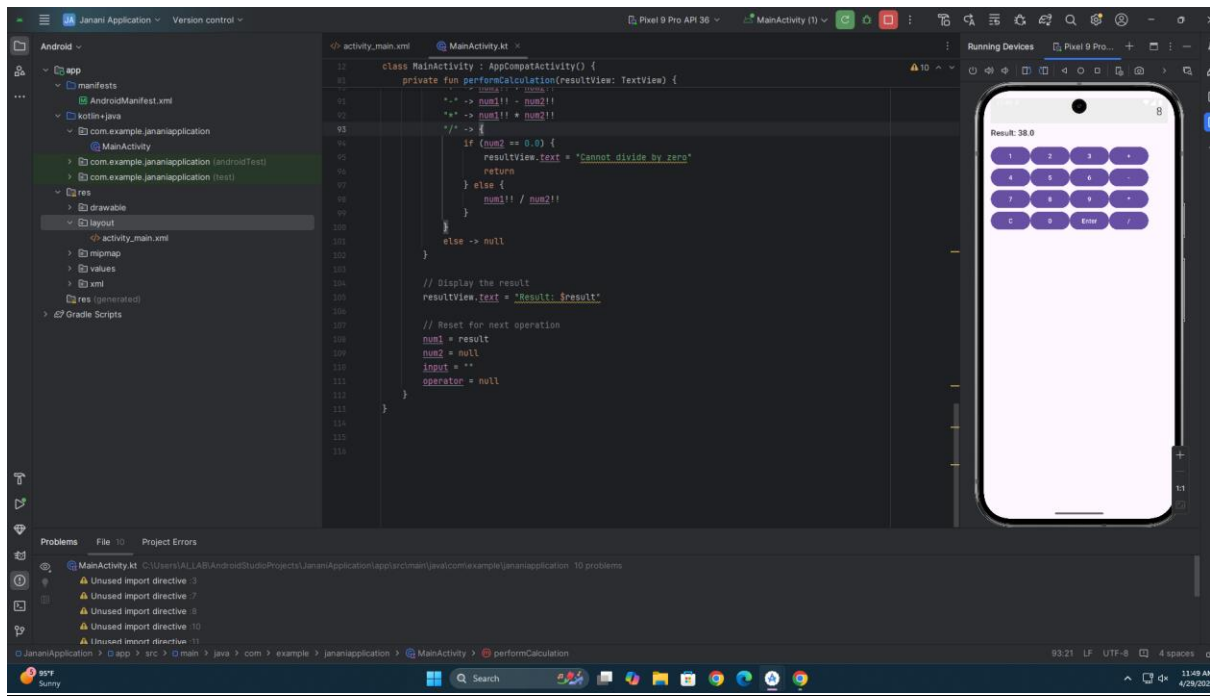
1  <?xml version="1.0" encoding="utf-8"?>
2  <androidx.constraintlayout.widget.ConstraintLayout
3      xmlns:android="http://schemas.android.com/apk/res/android"
4      xmlns:app="http://schemas.android.com/apk/res-auto"
5      android:layout_width="match_parent"
6      android:layout_height="match_parent"
7      android:padding="16dp">
8
9      <!-- Display Area -->
10     <TextView
11         android:id="@+id/tvInput"
12         android:layout_width="0dp"
13         android:layout_height="wrap_content"
14         android:text="0"
15         android:textSize="24sp"
16         android:gravity="end"
17         android:background="#EEEE"
18         android:padding="12dp"
19         app:layout_constraintTop_toTopOf="parent"
20         app:layout_constraintStart_toStartOf="parent"
21         app:layout_constraintEnd_toEndOf="parent" />
22
23     <!-- Result Display -->
24     <TextView
25         android:id="@+id/tvResult"
26         android:layout_width="0dp"
27         android:layout_height="wrap_content"
28         android:text="Result: "
29         android:textSize="18sp"
30         android:textStyle="bold"
31         android:layout_marginTop="8dp"
32         app:layout_constraintTop_toBottomOf="@id/tvInput"
33         app:layout_constraintStart_toStartOf="parent"
34         app:layout_constraintEnd_toEndOf="parent" />
35
36     <!-- Number Pad -->
37     <!-- Example row (1,2,3,+) -->
38     <GridLayout
39         android:id="@+id/gridPad"
40         android:layout_width="0dp"
41         android:layout_height="0dp"
42         android:layout_marginTop="16dp"
43         android:rowCount="4"
44         android:columnCount="4"
45         app:layout_constraintTop_toBottomOf="@id/tvResult"
46         app:layout_constraintBottom_toBottomOf="parent"
47         app:layout_constraintStart_toStartOf="parent"
48         app:layout_constraintEnd_toEndOf="parent">
49
50         <!-- Buttons 1-9, 0, C, Enter, +, -, *, / -->
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59         <Button android:text="8" android:id="@+id/btn8"/>
60         <Button android:text="9" android:id="@+id/btn9"/>
61         <Button android:text="0" android:id="@+id/btn0"/>
62         <Button android:text="C" android:id="@+id/btnClear"/>
63         <Button android:text="+" android:id="@+id/btnAdd"/>
64         <Button android:text="-" android:id="@+id/btnSubtract"/>
65         <Button android:text="*" android:id="@+id/btnMultiply"/>
66         <Button android:text="/" android:id="@+id/btnDivide"/>
67     </GridLayout>
68
69 </androidx.constraintlayout.widget.ConstraintLayout>

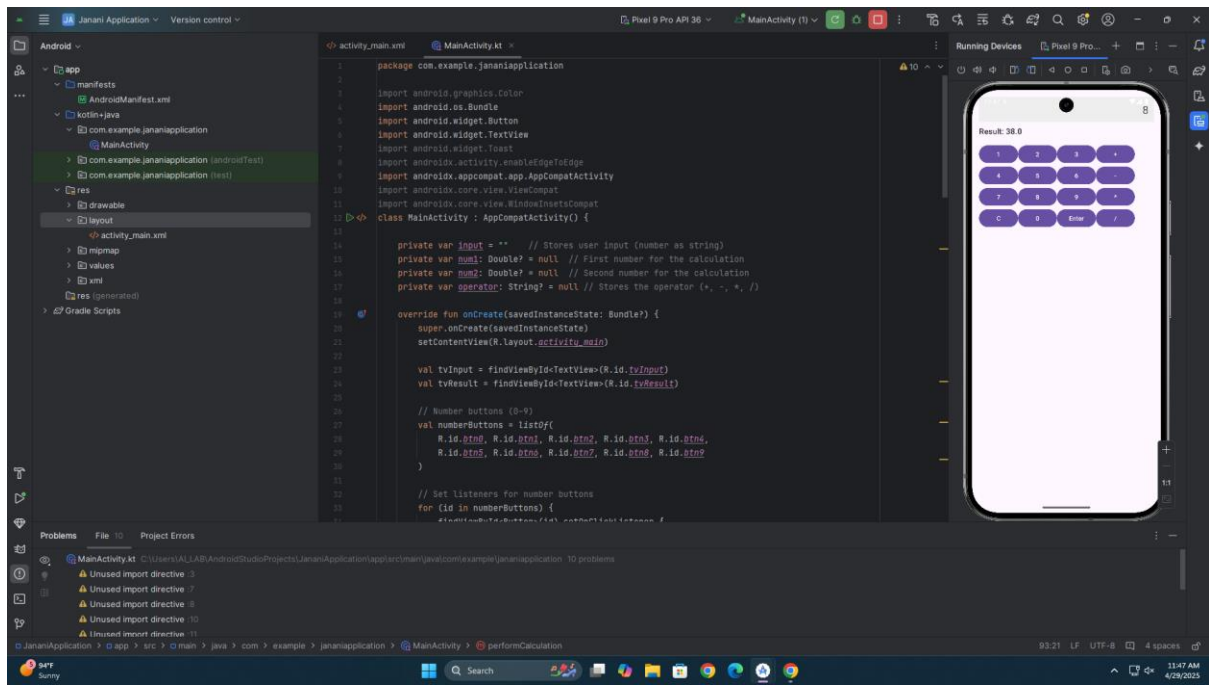
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44         android:columnCount="4"
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RESULT:

Thus the android application using controls like button, textView and edit text for designing a calculator having the basic functionalities like addition, subtraction, multiplication and division has been developed successfully.